



NORTH SYDNEY BOYS HIGH SCHOOL

2016 Year 11 Task 1

Mathematics Extension 1

General Instructions

- Reading time – **5 minutes**
- Working time – **80 minutes**
- Write on both sides of the lined paper in the booklet provided
- Write using blue or black **pen**
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Marks may not be awarded for illegible or incomplete working.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Mr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Mrs Ziazaris

Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	Total	Total %
Mark	<u>15</u>	<u>14</u>	<u>10</u>	<u>16</u>	<u>12</u>	<u>11</u>	<u>12</u>	<u>90</u>	<u>100</u>

Question 1 (15 marks)

Commence a NEW page

Mark

- (a) An atom of a certain metal weighs 7.31×10^{-26} grams.

2

How many atoms of this metal are in a 2 kilogram block?

Give your answer correct to three significant figures.

- (b) If $x = \sqrt{3} + 1$, find the value of $x^2 - \frac{1}{x^2}$ as a single fraction with a rational denominator.

3

- (c) Simplify $\frac{a^2 - a - 12}{a^2 - 9} \times \frac{6a - 18}{18a}$

2

- (d) Simplify $\frac{3^n + 6^n}{2 + 2^{n+1}}$

2

- (e) Factorise $6x^2 + 5xy - 4y^2$

2

- (f) Factorise $x^5 + 32$

2

- (g) Factorise $a^2 - b^2 + 2b - 1$

2

Question 2 (14 marks)

Commence a NEW page

Marks

Sketch graphs of the following curves, indicating any intercepts and asymptotes:

- (a) $y = \sqrt{2 - x^2}$

2

- (b) $y = (x^2 - 1)(x - 3)$

2

- (c) $y = 3 - \frac{1}{x - 2}$

3

- (d) $y = 2^{2-x}$

2

- (e) $y = \frac{x^2 + 3x - 10}{x - 2}$

2

- (f) Sketch $y = \sqrt{x+1}$ and hence sketch $y = \frac{1}{\sqrt{x+1}}$ showing any intercepts or asymptotes.

3

Question 3 (10 marks)

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Marks

(a) Write down the Domain and Range of these functions:

(i) $y = x^2 + 4x + 9$

2

(ii) $y = \frac{4}{\sqrt{25 - x^2}}$

2

(b) If $f(x) = \frac{x+5}{x-3}$, then(i) Find the equation of the inverse function $f^{-1}(x)$.

2

(ii) Find the coordinates of any points where the graphs of $y = f(x)$ and $y = f^{-1}(x)$ intersect.

2

(c) What is the natural domain of $f(x) = \sqrt{x-4} + \sqrt{10-x}$?

2

Question 4 (16 marks)

Commence a NEW page

Marks(a) Simplify $\tan(90^\circ - \theta)\sin(360^\circ - \theta)$, leaving your answer in terms of θ .

2

(b) Prove that $\cot A(\operatorname{cosec} A - \cot A) = \frac{\cos A}{1 + \cos A}$

3

(c) If $\sin \theta = \frac{1}{2}$ and $\tan \theta = \frac{-1}{\sqrt{3}}$, find θ for $0^\circ \leq \theta \leq 360^\circ$

2

(d) Solve these trigonometric equations:

(i) $\cos \theta = \sec \theta, 0^\circ \leq \theta \leq 360^\circ$

2

(ii) $\sin 2\theta = -\frac{1}{\sqrt{2}}, -180^\circ \leq \theta \leq 180^\circ$

3

(iii) $2\sin^2 \theta + 3\cos \theta = 0, 0^\circ \leq \theta \leq 360^\circ$

2

(iv) $2\cos(\theta - 30^\circ) = 0.7, 0^\circ \leq \theta \leq 360^\circ$

2

Question 5 (12 marks)

Commence a NEW page

Marks(a) Use about one-third of a page to sketch $y = 2 - 3\sin 2\theta$ for $0^\circ \leq \theta \leq 360^\circ$

3

(b) Bernie sails 30km from Port Cruz on a bearing of 300°T . Due to Cyclone Hillary he is then forced to change course and sail 80km on a bearing of 020°T to Trumpville.

(i) Sketch a diagram to show this information (be sure to show North!)

2

(ii) Calculate how far Trumpville is from Port Cruz, to the nearest km.

3

(c) A triangle has angles of size 45° , 55° , and 80° , and its longest side is 10cm long.

(i) Sketch this situation.

1

(ii) Show that the area A of this triangle is given by the formula

3

$$A = \frac{50 \sin 135^\circ \cos 35^\circ}{\cos 10^\circ}$$

Question 6 (11 marks)

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Mark

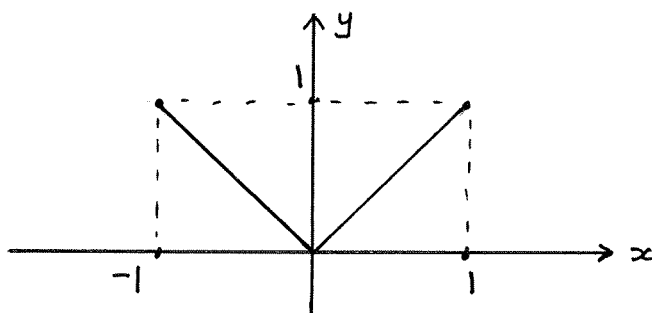
- (a) Solve $(x-4)^2 = 11$ 2
- (b) Solve $2x^2 + 3x - 5 > 0$ 2
- (c) Solve $|2x-1| < 5$ 2
- (d) Solve $|2x-1| = x-2$ 2
- (e) Solve $\frac{4x}{x-3} \leq 5$ 3

**Question 7** (12 marks)

Commence a NEW page

Mark

- (a) Sketch $y = \sqrt{x^2 - 2}$ 2
- (b) Solve $|x+1| + |x-3| = 7$ 2
- (c) Given that $x + y = 1$ and $x^3 + y^3 = 16$, evaluate $x^2 + y^2$. 3
- (d) Use the graph of $y = f(x)$ below to sketch the graph of $y = f(2x-4)$. 2



- (e) You are given that $f(x-1) + f(1-x) = 2x^2 + 8$.

Clearly, $2x^2 + 8$ is an even expression, since $2x^2 + 8$ equals $2(-x)^2 + 8$ for any real number x . But is the function $f(x)$ itself ODD, EVEN or NEITHER?

Justify your conclusion!

3

END OF EXAM

Q1

$$\begin{aligned}
 (a) \text{ Number atoms} &= \frac{2000}{7.31 \times 10^{-26}} \\
 &\div 2.73597811 \times 10^{28} \\
 &= 2.74 \times 10^{28} \quad (3 \text{ sig. figs.})
 \end{aligned}$$

$$(b) \quad x^2 = (\sqrt{3} + 1)^2 = 4 + 2\sqrt{3}$$

$$\therefore x^2 - \frac{1}{x^2} = 4 + 2\sqrt{3} - \frac{1}{4 + 2\sqrt{3}} \cdot \frac{4 - 2\sqrt{3}}{4 - 2\sqrt{3}}$$

$$= 4 + 2\sqrt{3} - \frac{4 - 2\sqrt{3}}{4}$$

$$= \frac{12 + 10\sqrt{3}}{4} \quad \left(= \frac{6 + 5\sqrt{3}}{2} \right)$$

$$(c) \quad \frac{a^2 - a - 12}{a^2 - 9} \times \frac{6a - 18}{18a} = \frac{(a-4)(\cancel{a+3})}{(\cancel{a+3})(a-3)} \cdot \frac{6(\cancel{a-3})}{18a}$$

$$= \frac{a-4}{3a}$$

$$(d) \quad \frac{3^n + 6^n}{2 + 2^{n+1}} = \frac{3^n(\cancel{1+2^n})}{2(\cancel{1+2^n})}$$

$$= \frac{3^n}{2}$$

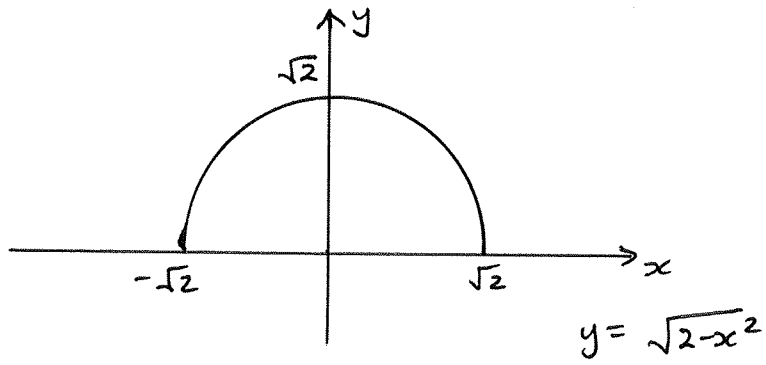
$$(e) \quad 6x^2 + 5xy - 4y^2 = (3x + 4y)(2x - y)$$

$$(f) \quad x^5 + 32 = (x+2)(x^4 - 2x^3 + 4x^2 - 8x + 16)$$

$$\begin{aligned}
 (g) \quad a^2 - b^2 + 2b - 1 &= a^2 - (b^2 - 2b + 1) \\
 &= a^2 - (b-1)^2 = (a+b-1)(a-b+1)
 \end{aligned}$$

Q2

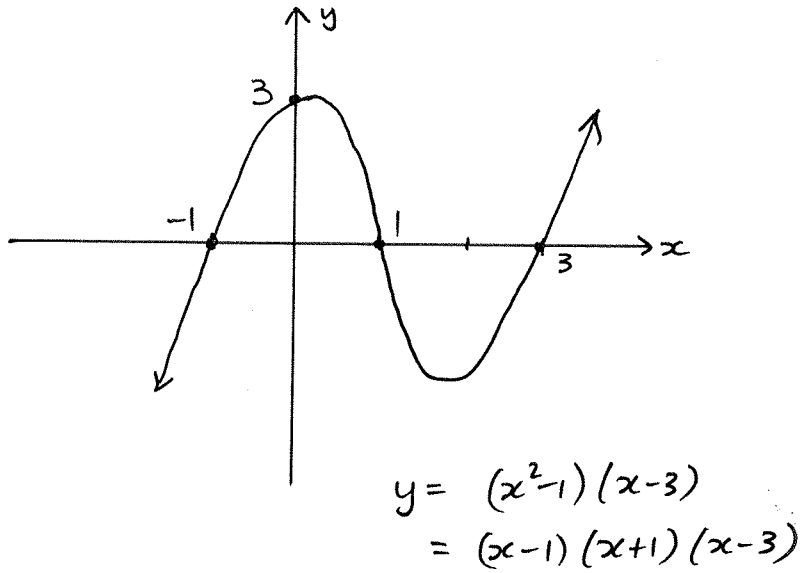
(a)



* Delete 1 mark for unlabelled axes (once).

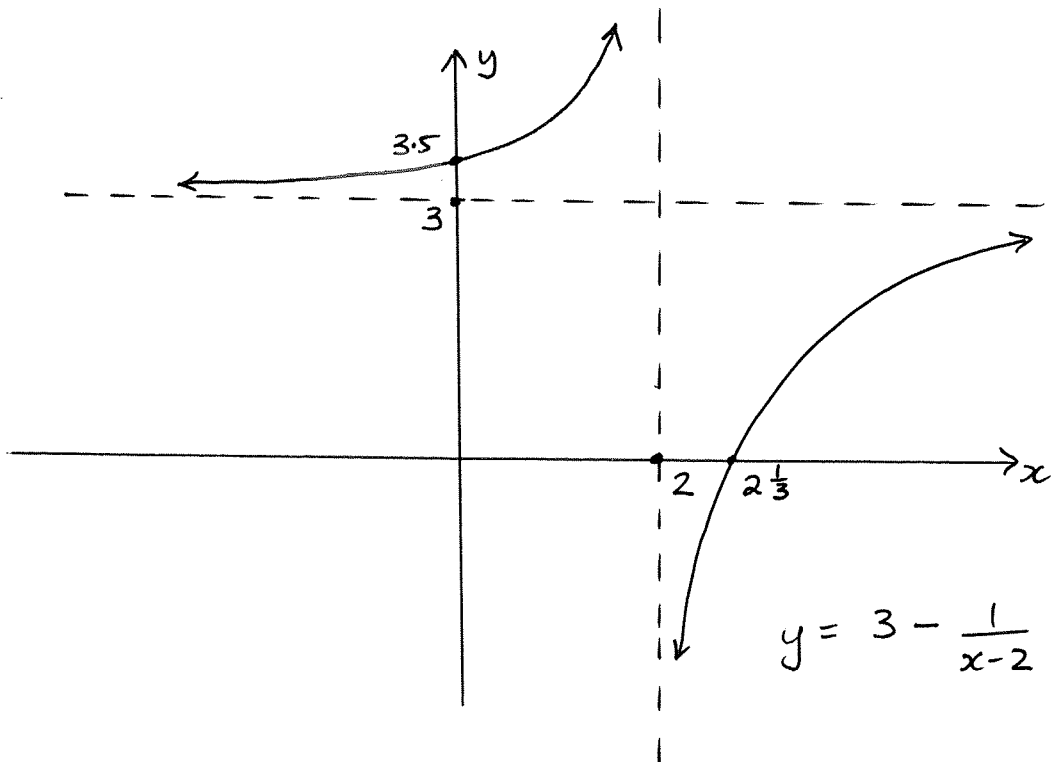
✓
✓

(b)



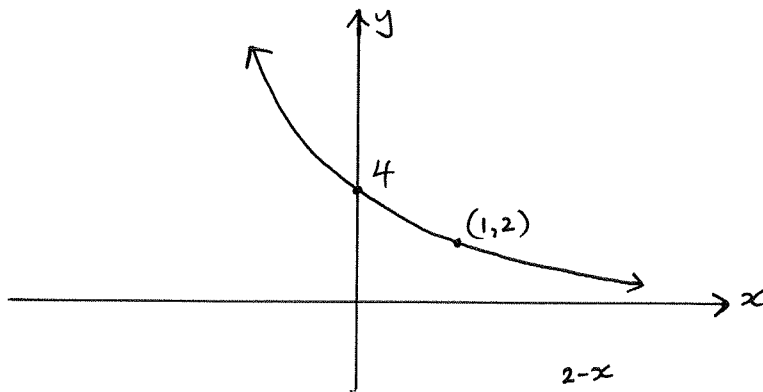
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(c)



✓
✓
✓

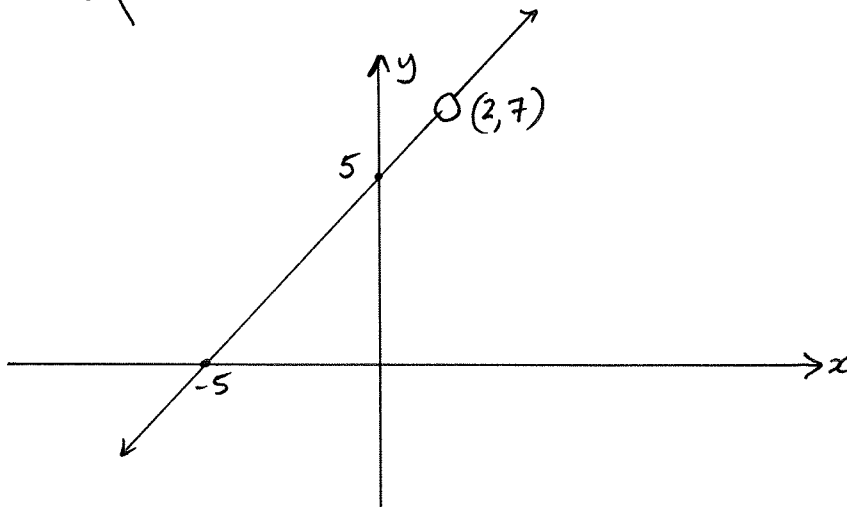
Q2 ctd
(d)



$$\begin{aligned} y &= 2^{2-x} \\ &= 2^2 \cdot 2^{-x} \\ &= 4 \cdot 2^{-x} \end{aligned}$$

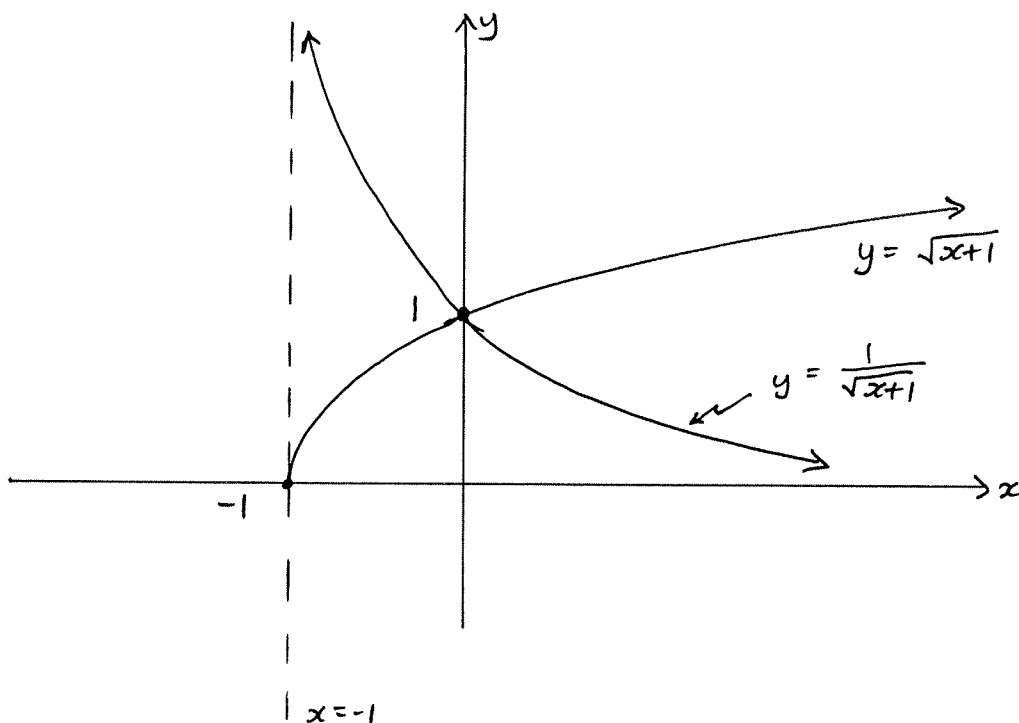
✓✓

(e) $y = \frac{(x+5)(x-2)}{x-2} \therefore y = x+5 \quad (x \neq 2)$



✓✓

(f)



✓

$\sqrt{x+1}$

✓✓

$\frac{1}{\sqrt{x+1}}$

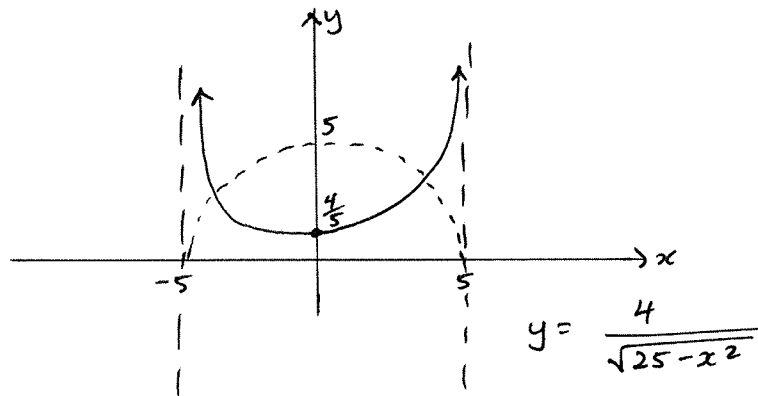
Q3 (a) (i) $y = x^2 + 4x + 9 = (x+2)^2 + 5$

$\therefore D: \text{all real } x$

$R: y \geq 5$

✓ D
✓ R

(ii)



$\therefore D: -5 < x < 5$

$R: y \geq \frac{4}{5}$

✓ D
✓ R

(b) $y = f(x) = \frac{x+5}{x-3}$

(i) For inverse, $x = \frac{y+5}{y-3}$

$\therefore xy - 3x = y + 5$

$(x-1)y = 3x+5$

$\therefore y = \frac{3x+5}{x-1}$

i.e. $f^{-1}(x) = \frac{3x+5}{x-1}$

✓
✓

(ii) $f(x)$ & $f^{-1}(x)$ intersect on $y = x$.

$\therefore$ solve simultaneously $y = x$ & $y = \frac{x+5}{x-3}$

$\therefore \frac{x+5}{x-3} = x \quad \therefore x^2 - 3x = x + 5$
 $x^2 - 4x - 5 = 0 \quad \therefore (x-5)(x+1) = 0$

$\therefore x = 5 \text{ or } -1 \quad \therefore \text{points are } (-1, -1), (5, 5).$

✓✓

(c) For $y = \sqrt{x-4}$, $D: x \geq 4$

For $y = \sqrt{10-x}$, $D: x \leq 10$

$\therefore$ For $y = \sqrt{x-4} + \sqrt{10-x}$, $D: 4 \leq x \leq 10$.

✓

✓

Q4

$$\begin{aligned}
 (a) \quad \tan(90-\theta) \cdot \sin(360-\theta) &= \cot \theta \cdot -\sin \theta \\
 &= \frac{\cos \theta}{\cancel{\sin \theta}} \cdot \cancel{-\sin \theta} \\
 &= -\cos \theta
 \end{aligned}$$

✓✓

$$\begin{aligned}
 (b) \quad \text{LHS} &= \cot A (\operatorname{cosec} A - \cot A) \\
 &= \frac{\cos A}{\sin A} \left(\frac{1}{\sin A} - \frac{\cos A}{\sin A} \right) \\
 &= \frac{\cos A (1 - \cos A)}{\sin^2 A} \\
 &= \frac{\cos A (1 - \cos A)}{1 - \cos^2 A} = \frac{\cos A (1 - \cancel{\cos A})}{(1 - \cancel{\cos A})(1 + \cos A)} \\
 &= \frac{\cos A}{1 + \cos A} = \text{RHS}
 \end{aligned}$$

✓

✓✓

$$\begin{aligned}
 (c) \quad &\begin{cases} \sin > 0 \text{ and } \tan < 0 \therefore \theta \text{ is in Q2} \\ \text{related angle is } 30^\circ \end{cases} \\
 &\therefore \theta = 150^\circ
 \end{aligned}$$

✓✓

$$\begin{aligned}
 (d) \quad (i) \quad \cos \theta &= \sec \theta, \quad 0^\circ \leq \theta \leq 360^\circ. \\
 &= \frac{1}{\cos \theta}
 \end{aligned}$$

$$\therefore \cos^2 \theta = 1$$

$$\therefore \cos \theta = \pm 1$$

$$\therefore \theta = 0^\circ, 180^\circ, 360^\circ.$$

✓✓

Q4 - ctd

$$(d) (ii) \quad \sin 2\theta = -\frac{1}{\sqrt{2}}, \quad -180^\circ \leq \theta \leq 180^\circ$$

$$\therefore -360^\circ \leq 2\theta \leq 360^\circ$$

Related angle is 45° , Quadrants are 3 & 4.

$$\therefore 2\theta = 225^\circ, 315^\circ, -45^\circ, -135^\circ$$

$$\theta = 112.5^\circ, 157.5^\circ, -22.5^\circ, -67.5^\circ$$

✓✓✓

$$(iii) \quad 2 \sin^2 \theta + 3 \cos \theta = 0, \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\therefore 2(1 - \cos^2 \theta) + 3 \cos \theta = 0, \quad 2 - 2 \cos^2 \theta + 3 \cos \theta = 0$$

$$\therefore 2 \cos^2 \theta - 3 \cos \theta - 2 = 0$$

$$\therefore (2 \cos \theta + 1)(\cos \theta - 2) = 0$$

$$\therefore \cos \theta = -\frac{1}{2} \quad \text{or} \quad \cos \theta = 2$$

↑
not possible

$$\therefore \theta = 120^\circ, 240^\circ$$

✓✓

$$(iv) \quad 2 \cos(\theta - 30^\circ) = 0.7, \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\therefore \cos(\theta - 30^\circ) = 0.35, \quad -30^\circ \leq \theta - 30^\circ \leq 330^\circ$$

Related angle is $69^\circ 31'$, in Q1 or Q4

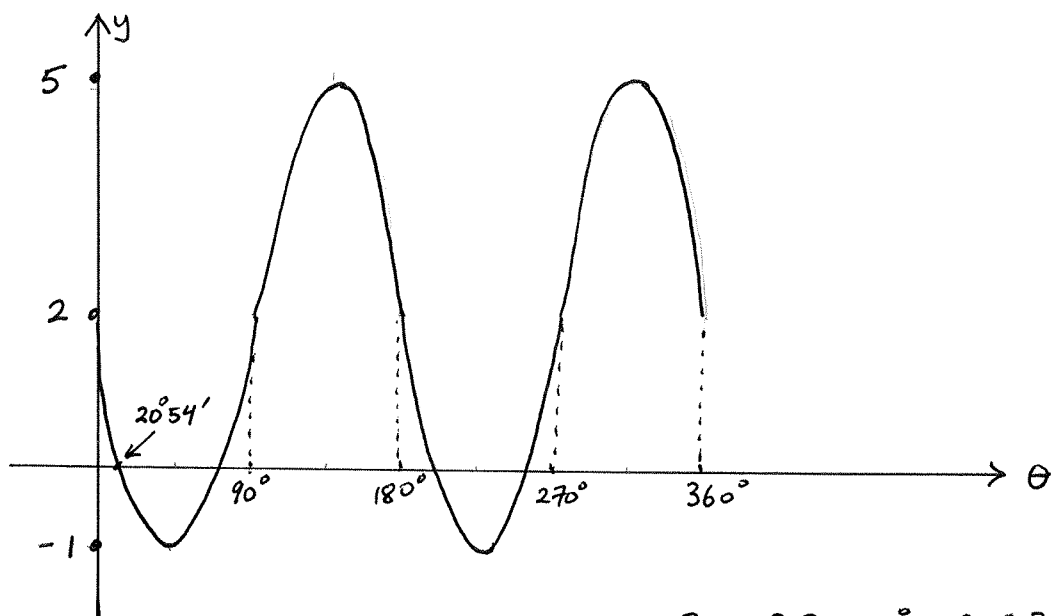
$$\therefore \theta - 30^\circ = 69^\circ 31' \text{ or } 290^\circ 29'$$

$$\therefore \theta = 99^\circ 31' \text{ or } 320^\circ 29'$$

✓✓

Q5

(a)

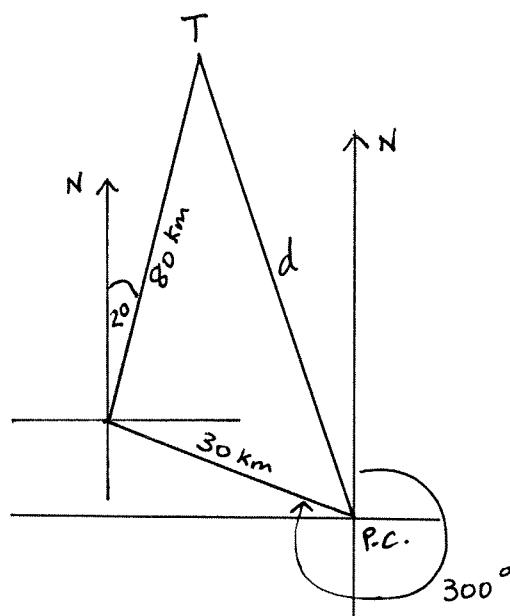


$$y = 2 - 3 \sin 2\theta, \quad 0^\circ \leq \theta \leq 360^\circ$$

✓✓✓

(b)

(i)



(* unlabelled
North →
deduct 1 mk.)

✓✓

(ii) Large angle in triangle = $70^\circ + 30^\circ = 100^\circ$

$$\therefore d^2 = 30^2 + 80^2 - 2(30)(80) \cos 100^\circ$$

$$\div 8133.511253$$

$$\therefore d \div 90.18598$$

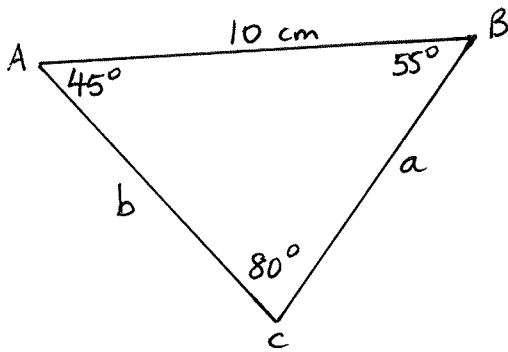
$$\therefore \text{distance} = 90 \text{ km (nearest km)}$$

✓
✓

✓

Q5-ctd

(c)



(i)

(ii) We use sine rule to find one other side, then use the area formula.

$$\text{eg. } \frac{a}{\sin 45^\circ} = \frac{10}{\sin 80^\circ}$$

$$\therefore a = \frac{10 \sin 45^\circ}{\sin 80^\circ}$$

$$\text{Thus area} = \frac{1}{2} \cdot 10 \cdot a \cdot \sin 55^\circ$$

$$= \frac{1}{2} \cdot 10 \cdot \frac{10 \sin 45^\circ}{\sin 80^\circ} \cdot \sin 55^\circ$$

$$= 50 \frac{\sin 45^\circ \sin 55^\circ}{\sin 80^\circ}$$

$$\therefore \text{area} = \frac{50 \sin 135^\circ \cos 35^\circ}{\cos 10^\circ}, \text{ as required.}$$

(since $\sin \theta = \sin (180 - \theta)$ and $\sin \theta = \cos (90 - \theta)$).

Q6

$$(a) \quad (x-4)^2 = 11$$

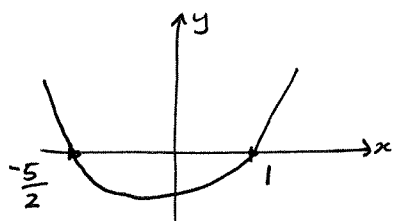
$$\therefore x-4 = \pm \sqrt{11}$$

$$\therefore x = 4 + \sqrt{11} \quad \text{or} \quad 4 - \sqrt{11}$$

✓✓

$$(b) \quad 2x^2 + 3x - 5 > 0$$

$$(2x+5)(x-1) > 0$$



$$\therefore x < -\frac{5}{2} \quad \text{or} \quad x > 1$$

✓✓

$$(c) \quad |2x-1| < 5$$

$$\therefore -5 < 2x-1 < 5$$

$$\therefore -4 < 2x < 6$$

$$\therefore -2 < x < 3$$

✓✓

$$(d) \quad |2x-1| = x-2$$

$$\therefore 2x-1 = x-2$$

$$x = -1$$

$$\text{or} \quad 2x-1 = -x+2$$

$$3x = 3$$

$$x = 1$$

$$\therefore x = -1 \quad \text{or} \quad x = 1$$

✓

Test: $x = -1 \rightarrow$ LHS = $|-2-1| = 3$
 RHS = $-1-2 = -3$ $\times$ not a solution

$x = 1 \rightarrow$ LHS = $|2-1| = 1$
 RHS = $1-2 = -1$ $\times$ not a solution

$\therefore$ no solutions.

✓

Q6-cld

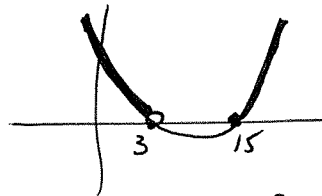
$$(e) \frac{4x}{x-3} \leq 5 \quad (\because x \neq 3)$$

$$\therefore 4x(x-3) \leq 5(x-3)^2$$

$$\therefore 5(x-3)^2 - 4x(x-3) \geq 0$$

$$(x-3) [5(x-3) - 4x] \geq 0$$

$$(x-3)(x-15) \geq 0$$

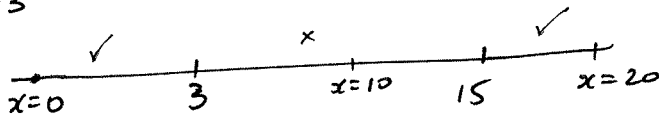


$$\therefore x < 3 \text{ or } x \geq 15$$

[ALT.1 (Critical point method)]

$x \neq 3 \therefore x = 3$ is critical point.

$\frac{4x}{x-3} = 5 \Rightarrow 4x = 5x - 15 \therefore x = 15$ is critical point.



Testing a point in each region,

eg. at $x=0 \rightarrow \frac{0}{-3} = 0 < 5 \therefore \checkmark$

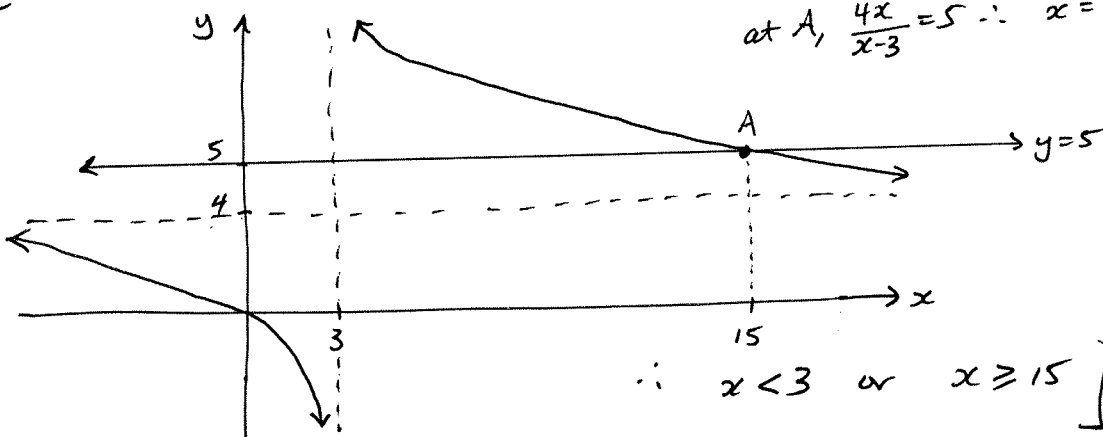
at $x=10 \rightarrow \frac{40}{7} > 5 \therefore \times$

at $x=20 \rightarrow \frac{80}{17} < 5 \therefore \checkmark$

Thus $x < 3$ or $x \geq 15$

[ALT.2 (Graphical) $y = \frac{4x}{x-3} = \frac{4(x-3)+12}{x-3} = 4 + \frac{12}{x-3}$

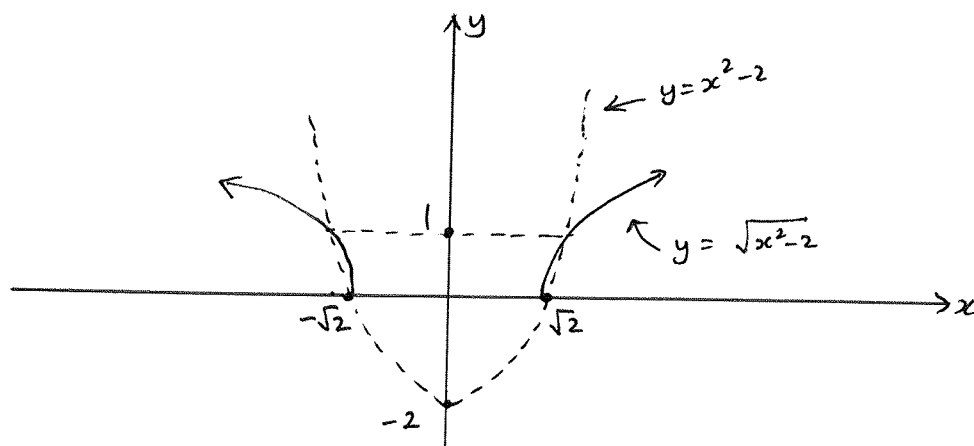
at A, $\frac{4x}{x-3} = 5 \therefore x = 15$



$\therefore x < 3$ or $x \geq 15$

Q7

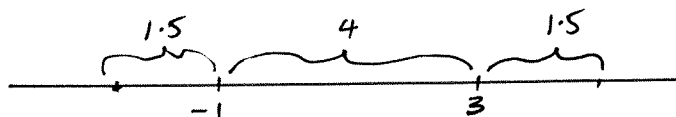
(a)



✓✓

(b) $|x+1| + |x-3| = 7$

On a numberline, the distance of x from -1 plus the distance of x from 3 equals 7 :

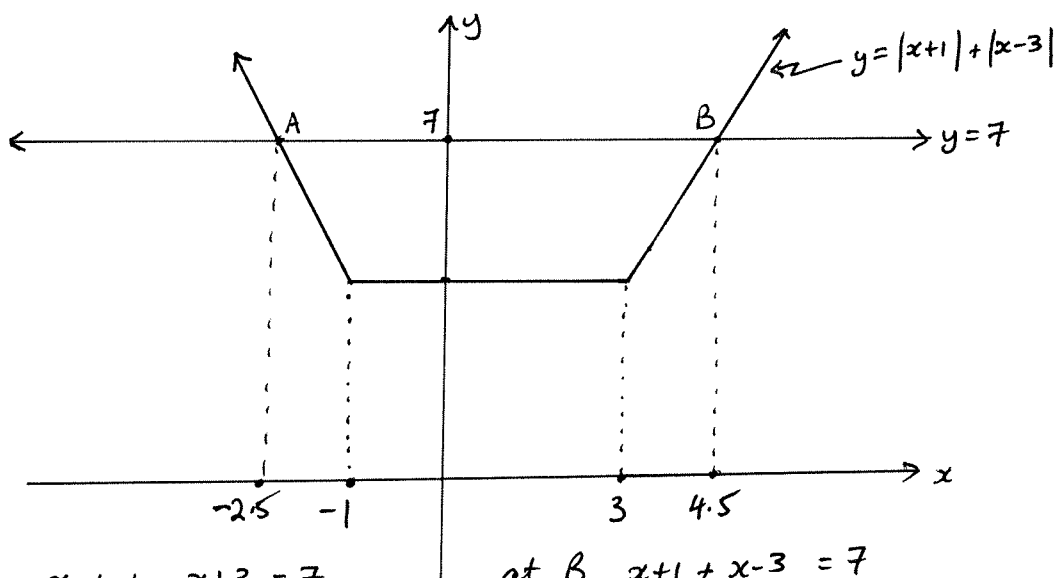


This will be satisfied by points 1.5 units to the right of $x=3$ or 1.5 units to the left of $x=-1$.

$\therefore x = 4.5$ or $x = -2.5$

✓✓

[ALT.]



at A, $-x-1 + -x+3 = 7$
 $-2x = 5$
 $x = -2.5$

at B, $x+1 + x-3 = 7$
 $2x = 9$
 $x = 4.5$

$\therefore x = -2.5$ or 4.5 .]

Q7-cld

(c) $x+y=1$ and $x^3+y^3=16$. Find x^2+y^2 .

$$x^3+y^3 = (x+y)(x^2-xy+y^2)$$

$$\therefore 16 = 1(x^2+y^2-xy)$$

$$\therefore x^2+y^2 = 16+xy$$

$$\text{But } (x+y)^2 = x^2 + 2xy + y^2$$

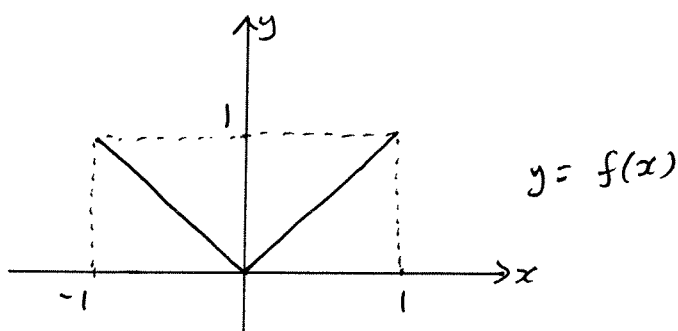
$$1^2 = x^2+y^2 + 2xy$$

$$\therefore x^2+y^2 = 1-2xy$$

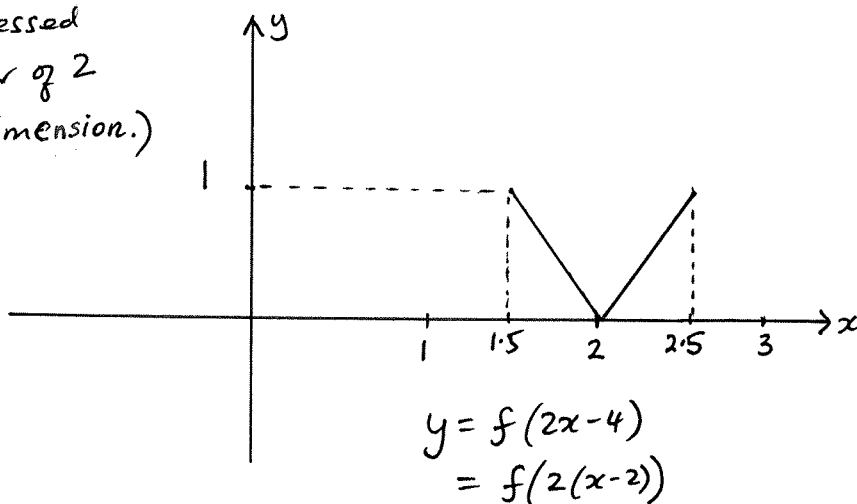
$$\therefore 16+xy = 1-2xy \quad \therefore xy = -5$$

$$\text{Thus } x^2+y^2 = 16-5 = 11$$

(d)



(The graph has been shifted 2 units to the right,
and compressed
by a factor of 2
in the x dimension.)



Q7-ctd

$$(e) \quad f(x-1) + f(1-x) = 2x^2 + 8$$

$f(x)$ has to be odd, or even, or neither.

(no other possibilities!)

IF $f(x)$ is odd, then $f(-x) = -f(x)$.

$$\therefore f(x-1) + f(1-x) = f(x-1) - f(x-1) = 0, \text{ for all } x$$

But $2x^2 + 8 \neq 0$ for any x .

$\therefore f(x)$ is not odd. ✓

IF $f(x)$ is even, then

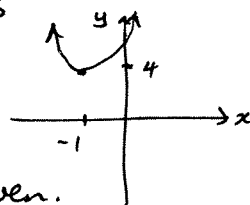
$$f(x-1) + f(1-x) = 2 \cdot f(x-1)$$

$$\therefore f(x-1) = x^2 + 4$$

$$\therefore f(x) = (x+1)^2 + 4$$

$$= x^2 + 2x + 5$$

But this is not even! viz:



Contradiction! $\therefore f(x)$ is not even. ✓

So if $f(x)$ cannot be odd, and cannot be even, ✓

$\therefore$ it must be neither. ✓